

The early challenges

Like with any other large scale project, the development of a huge ePDS also faced some challenges initially. As Ashish Nauriyal, a lead solutions consultant at EnterpriseDB, explained in a conversation with *Open Source For You*, the major challenges were designing, deploying and maintaining Postgres for a mission critical, statewide ePDS. "Some of the specific challenges faced were appropriate Postgres versions and project-relevant features, database and server-level tuning, expert guidance, implementation of best practices, tools required to achieve desired database-level tasks such as diagnosis of performance issues, monitoring, high-availability, maintenance operations, and ongoing database server and application level changes," recalls Nauriyal.

The engineers leveraged their extensive experience in the database management space to overcome these challenges. The engineering team provided support for disaster recovery, security controls, performance tuning, and backups and upgrades in addition to designing and implementing systems. Also, the NIC team was allowed to change the management for the database with minimal or even no production database service disruption.

For the team at NIC, the major obstacle was the execution of tasks like biometric authentication, the Aadhaar-enabled payment interface and hyper-scaling. "With the appropriate architecture and optimisation through the efforts of EnterpriseDB, the solution worked out flawlessly," says Rajasekhara.

However, as the state government was committed to improving transparency and efficiency, no major administrative challenges were faced in opting for the advanced ePDS model.

Linux as a software stack

The NIC engineers used Linux as the software stack for the Postgres deployment. This mix is considered to be the real icing on the cake. "Postgres and Linux originate from a like-minded

Aadhaar enabled Public Distribution System (AePDS)
Department of Consumer Affairs, Food & Civil Supplies
Government of Andhra Pradesh

Select Date: 15

Regular: PDS
Date: 16-05-2017

S. No.	District Name	Shop	Cards	Units	Trans	Issued	Cancelled	Active	Inactive	Portability	Available	Total	Cashless	Cash	Issued
1	Krishna	1262	129618	261234	188	39	32	2180	17778	194326	8422	82686	82278	294346	
2	West Godavari	1343	199418	328466	15	15	2	15	2140	17315	197843	8941	8941	5126	29538
3	Srikantam	1975	499402	2122247	27	27	3	19	1911	29111	71222	8720	24621	46127	269966
4	Kurum	2622	116979	362286	107	107	5	32	2281	17813	94336	8236	27081	28127	269586
5	East Godavari	1341	149869	466763	47	47	4	24	2417	26218	136866	8431	34936	26108	34231
6	YSR Kothapeta	1728	70367	216476	17	17	6	26	1706	42611	64811	8535	12246	18197	120365
7	Guntur	2742	143219	854229	21	21	6	17	2732	28471	116108	8181	15479	16593	320747
8	Chittoor	2648	198019	227941	23	23	6	16	2832	27942	94820	8839	10719	14584	298613
9	Prakasam	2115	101991	284040	22	22	1	17	2108	9710	78020	8316	9884	12556	224715
10	Vidyanagar	1346	47776	119162	15	15	6	30	1316	2646	44577	8545	11329	17128	
11	Anantapur	2642	121014	366466	14	14	43	3991	8482	381033	8612	7664	7227	320000	
12	Narayan	1889	60307	221761	36	36	6	26	1863	43211	71660	8284	30717	5532	192761
13	Vishakhapatnam	1695	149819	289981	76	76	6	36	16769	389220	8616	36518	4059	225866	
	Grand Total	20616	1386912	2691573	723	723	52	267	28248	143273	11040475	8615	3162433	2646	3481573

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The ePDS project: Maintaining humongous public distribution records in a digital form

Techniques used to secure data in EnterpriseDB Postgres

- **Authentication:** Implementation of advanced levels of authentication such as SSPI, LDAP and Kerberos
- **Authorisation:** Object level privileges to allow or revoke user access
- **Auditing:** Tracking and analysing database activities to ensure data security
- **Encryption:** Password storage encryption, encrypted columns, password protection on data across networks and SSL host authentication
- **Prevention from SQL injection attacks:** SQL/Protect deployed to provide a layer of security to give control back to the database administrator whenever potentially dangerous queries are received

community. This community essence makes Linux an optimal platform for running Postgres," says Nauriyal.

The presence of Linux enables experts like Nauriyal to control the database with ease. It also provides the flexibility to choose from multiple file systems to enable satisfactory results.

How EnterpriseDB helped in the deployment of the ePDS model

"EnterpriseDB included low-CAPEX and OPEX in the solution they gave us," says Rajasekhara. EnterpriseDB's engineers helped NIC pilot the ePDS model for a district for two months, and subsequently scaled this up to the entire state. Application server clusters with disaster recovery (DR) and business continuity (BC) architecture were used to run the Postgres database on a private cloud.

"Control and visibility of a platform, along with the flexibility to choose from multiple file systems, help database administrators to manage a Postgres database that is performing well," says Ramakrishna Malempati, senior database consultant, EnterpriseDB.

Other open source deployments

Apart from PostgreSQL, the NIC team also deployed Cassandra and MongoDB as the database solutions to make its ePDS, eSCM and ePOS successful in the state. The use of all these open source solutions has reduced the CAPEX and OPEX of this project to zero.

The success of AP's ePDS model has led to other states planning for its replication, with Rajasthan and Maharashtra being the first two states to do so. 

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